Appendix: AI Usage

Sum Yee Chong

Table of contents

AI Usage 1

AI Usage

I used ChatGPT to support my understanding and improve the quality of this assignment. The AI was mainly used to clarify key concepts (such as reproducible reports), assist with debugging R code, and improve the clarity of my written explanations.

For example, I asked questions about data loading issues, handling missing values, and how to correctly use functions such as `filter()`, `count()`, and `group_by()`. I also used ChatGPT to understand how to apply inline R code and structure the report according to the assignment requirements.

In addition, I used ChatGPT to refine the wording of my introduction, dataset description, results, and conclusion sections to ensure clarity and appropriate academic tone. All analysis, interpretation, and final decisions were completed independently by me.

Screenshots of relevant interactions are included in the appendix as evidence of how AI was used in this assignment.

A full record of my AI interaction is available here: https://chatgpt.com/share/69d73441-6794-839a-960d-7db6a4caf882

I have an assignment about analysing the relationship between marketing spend and purchase amount in R. I'm not fully sure what "reproducible report" means in practice. Could you explain it in simple terms?

A **reproducible report** just means that *someone else (or future you) can run your work and get the same results*—without guessing what you did.

Think of it like this:
👉 Instead of only showing your final graphs or conclusions, you also show **the exact steps (code + data + explanation)** used to get there.

🔍 Simple explanation

A reproducible report combines three things in one place:

1. **Code** (what you ran in R)
2. **Output** (tables, graphs, results)
3. **Explanation** (what it means)

So if your tutor clicks "Run", everything updates automatically and gives the same results.

📊 Example (your assignment context)

You're analysing:

> marketing spend → purchase amount

A reproducible report would include:

1. Load data

```R
```

2. Analyse relationship

```R
```

3. Plot it

```R
```

4. Explain in words

> "The positive slope suggests that higher marketing sper
> purchase amounts."

ChatGPT

🧠 Why this matters

- ✅ Your tutor can **verify your work**
- ✅ No "how did you get this?" confusion
- ✅ Easy to fix or update (change data → results update

🛠 How you usually make one in R

Most likely using:

- **R Markdown** (.Rmd file)
- or **Quarto**

These let you mix:

- text
- code chunks
- graphs

Then "knit" into:

- HTML / PDF / Word report

💡 One-line summary

A reproducible report =
👉 *"Here is my analysis, AND here is exactly how I did it so you can reproduce it."*

If you want, I can help you **structure your actual assignment (intro → method → results → discussion)** or even give you a template you can paste into R Markdown.

```
Error in `filter()`:
ℹ In argument: `!is.na(purchase_amount)`.
Caused by error:
! object 'purchase_amount' not found
▼ Hide Traceback
```

This error is very common in R—good news is it's usually easy to fix 👍

❗ What the error means

```
object 'purchase_amount' not found
```

👉 R is telling you:

> "I can't find a column called `purchase_amount` in your data."

So when you run:

```R
filter(!is.na(purchase_amount))
```

R doesn't know where `purchase_amount` is coming from.

🔍 Most likely reasons (and fixes)

✅ 1. Column name is wrong

Maybe your dataset doesn't have a column exactly called `purchase_amount`.

👉 Check column names:

```R
colnames(data)
```

Common issues:

- `Purchase_Amount` (capital letters)
- `purchase amount` (space instead of underscore)
- `purchaseAmount` (camelCase)

👉 Fix:

```R
filter(!is.na(Purchase_Amount))  # match EXACT name
```

✅ 2. You didn't specify the dataset

If you're using **dplyr**, `filter()` needs a dataset.

Wrong:

```R
filter(!is.na(purchase_amount))
```

Correct:

```R
data %>% filter(!is.na(purchase_amount))
```

OR:

```R
filter(data, !is.na(purchase_amount))
```

✅ 3. Dataset not loaded / named differently

Maybe your data isn't called `data`.

👉 Check what objects exist:

```R
ls()
```

If your dataset is called `marketing_data`, then:

```R
marketing_data %>% filter(!is.na(purchase_amount))
```

✅ 4. Column name has spaces

If your column is like `"Purchase Amount"`:

👉 Use backticks:

```R
filter(!is.na(`Purchase Amount`))
```

🧠 Quick checklist

Before running `filter()`:

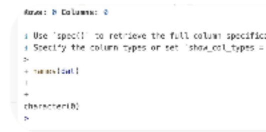
- ✔ Did I load the dataset?
- ✔ Am I using the correct dataset name?
- ✔ Is the column name EXACT?
- ✔ Am I piping (`%>%`) correctly?

💡 Example (correct version)

```R
library(dplyr)

clean_data <- data %>%
  filter(!is.na(purchase_amount))
```

If you want, paste your `colnames()` **output or code**, and I'll pinp
wrong in your case 👌



Ahh this screenshot actually explains the *real problem* 👇

❗ What's going wrong

```R
Rows: 0 Columns: 0
names(dat)
character(0)
```

👉 This means:

> Your dataset `dat` is **completely empty** (no rows, no columns

So of course R says:

```R
object 'purchase_amount' not found
```

because there are literally **no columns at all**.

🧠 Root cause (important)

This is **NOT a filter() problem** ❌
This is a **data loading problem** ⚠️

Your file didn't load properly.

🔍 Most common reasons + fixes

✅ 1. File path is wrong

If you used something like:

```r
dat <- read_csv("marketing.csv")
```

👉 R might not be finding the file.

Check:

```r
getwd()   # where R is looking
```

Fix:

```r
dat <- read_csv("C:/Users/yourname/Desktop/marketing.csv")
```

✅ 2. File is empty or wrong file

Open your CSV:

- Does it actually have data?
- Or is it blank?

✅ 3. Wrong delimiter (VERY common)

If your file uses `;` instead of `,`, **readr** may fail silently.

Try:

```r
dat <- read_csv2("marketing.csv")
```

✅ 4. File encoding / format issue

Sometimes Excel files saved weirdly → R reads nothing.

Fix:

- Re-save as **CSV UTF-8**
- Then reload

✅ 5. You overwrote your data accidentally

Maybe earlier you did:

```r
dat <- data.frame()
```

👉 That would wipe everything.

🧪 What to do RIGHT NOW

Run these in order:

```r
# 1. Reload your data
dat <- read_csv("your_file.csv")

# 2. Check structure
str(dat)

# 3. Check names
names(dat)

# 4. Preview
head(dat)
```

💡 What you should see (correct)

```r
Rows: 100 Columns: 3
[1] "marketing_spend" "purchase_amount"
```

🚨 **Key takeaway**

👉 Your error is happening because:

> "Your dataset is empty, not because your code is wrong."

If you send me:

- your `read_csv()` line
- or your CSV file screenshot

I can fix it with you step-by-step 👍

In the excel there is a column call segment how to check all the segments type with code

Good question—this is exactly the kind of thing you should show in a reproducible report 👍

If you have a column called `segment`, here are the main ways to **check all segment types in R**:

✅ 1. See all unique segment values

This shows *what kinds of segments exist*:

```R
unique(dat$segment)
```

👉 Example output:

```
[1] "A" "B" "C"
```

✅ 2. Count how many in each segment (MOST useful)

This is usually what your assignment wants:

```R
table(dat$segment)
```

👉 Example:

```
A   B   C
50  30  20
```

✅ 3. Using dplyr (cleaner for reports)

```R
library(dplyr)

dat %>%
  count(segment)
```

👉 Output:

```
segment    n
A          50
B          30
C          20
```

✅ 4. Check the data type of the column

This tells you if it's categorical (factor) or text:

```R
str(dat$segment)
```

or

```R
class(dat$segment)
```

💡 Bonus (very useful for your assignment)

If you want to **see segments + average purchase amount**:

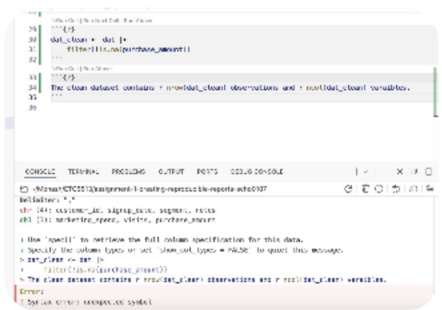
```R
dat %>%
  group_by(segment) %>%
  summarise(avg_purchase = mean(purchase_amount, na.rm = TRUE))
```

👉 This directly links to your topic:
marketing → purchase behaviour by segment

🧠 One-line summary

- `unique()` → what segments exist
- `table()` / `count()` → how many in each
- `group_by()` → deeper analysis

If you want, I can help you turn this into a **nice table + explanation paragraph** for your assignment (so you get better marks).



From your screenshot, I can see this line:

```R
dat_clean <- dat |>
```

👉 The issue is: **your code is incomplete after the pipe (** `|>`
yet.

✅ How to check `segment` properly (fix your cod

You need to continue the pipeline. For example:

✓ Count each segment

```R
dat_clean <- dat |>
  count(segment)
```

✓ Or just check segments without saving

```R
dat |>
  count(segment)
```

✓ If you want to clean first, then count

```R
dat_clean <- dat |>
  filter(!is.na(segment)) |>
  count(segment)
```

❗ Common mistake (you might be doing this)

```R
dat_clean <- dat |>
```

➡️ This alone does nothing → R is waiting for the next step

🧠 If `count()` doesn't work

Make sure you loaded **dplyr**:

```R
library(dplyr)
```

💡 Alternative (base R)

If you want something simple:

```R
table(dat$segment)
```

🔍 If you get another error

Run this to double check column name:

```R
names(dat)
```

👉 Maybe it's:

- `Segment`
- `segment_type`
- or something similar

✅ Best version for your assignment

```R
library(dplyr)

dat |>
  count(segment) |>
  arrange(desc(n))
```

👉 This shows:

- all segment types
- sorted from most common → least

If you send me your `names(dat)` output, I can **confirm exact**
called (sometimes it's slightly different and causes errors).

Ahh perfect—this is about **inline code in R Markdown**, and yo
shows the exact mistake 👍

❗ Your current error

```R
The clean dataset contains r nrow(dat_clean) observ
```

👉 Problem:

- You wrote `r nrow(...)` as **normal text**
- R Markdown doesn't recognize it → causes **syntax error**

✅ Correct inline code syntax

You must wrap it with backticks:

```R
`r ...`
```

✅ Fixed version

```R
The clean dataset contains `r nrow(dat_clean)` obse
```

👉 When you knit, it will show something like:

://chatgpt.com/share/69d73441-6794-839a-960d-7db6a4caf882

> The clean dataset contains **100 observations and 5 variables.**

🧠 Key idea (super important)

- Code chunk → runs blocks of R code
- Inline code → inserts **small results inside sentences**

✨ More examples (useful for your assignment)

✔ Number of segments

```R
There are `r length(unique(dat$segment))` customer segments.
```

✔ Most common segment

```R
The most common segment is `r names(sort(table(dat$segment), decreasing =
```

✔ Average purchase amount

```R
The average purchase amount is `r round(mean(dat$purchase_amount, na.rm =
```

🚨 Common mistakes to avoid

❌ Missing backticks
❌ Using quotes `"r nrow(...)"`
❌ Running inline code inside code chunk

💡 Pro tip (for higher marks)

Use inline code to make your report:

- dynamic
- professional
- reproducible

👉 Tutors LOVE this

If you want, I can help you write a **full paragraph using inline**
purchase analysis (very useful for your report).

Great question—this is exactly where many people get confus

❗ Short answer

👉 **Inline code is NOT inside a code chunk**

🧠 Difference (very important)

✅ Code chunk

Used for **running multiple lines of R code**

```Markdown
```{r}
dat_clean <- dat |>
  filter(!is.na(purchase_amount))
```
```

👉 Output appears **separately** (tables, plots, etc.)

✅ Inline code

Used **inside normal sentences**

```Markdown
```

Use inline code to make your report:

- dynamic
- professional
- reproducible

👉 Tutors LOVE this

If you want, I can help you write a **full paragraph using inline code** for your marketing vs purchase analysis (very useful for your report).

So is it inside code chunk

Great question—this is exactly where many people get confused 👍

❗ Short answer

👉 **Inline code is NOT inside a code chunk**

🧠 Difference (very important)

✅ Code chunk

Used for **running multiple lines of R code**

Markdown

```
```{r}
dat_clean <- dat |>
  filter(!is.na(purchase_amount))
```
```

👉 Output appears **separately** (tables, plots, etc.)

✅ Inline code

Used **inside normal sentences**

Markdown

```
The dataset contains `r nrow(dat_clean)` obser
```

👉 Output appears **inside the sentence**

🔍 Think of it like this

- Code chunk = "do the work" 🔧
- Inline code = "report the result nicely" 📝

✅ How they work together (your assignmen

Step 1: Code chunk (create data)

Markdown

```
```{r}
dat_clean <- dat |>
  filter(!is.na(purchase_amount))
```
```

Step 2: Inline code (describe result)

Markdown

```
The clean dataset contains `r nrow(dat_clean)`
```

❌ What NOT to do

Putting inline code inside a chunk:

Markdown

```
```{r}
The dataset has `r nrow(dat_clean)`
```
```

👉 This will give errors ❌

💡 Simple rule to remember

👉 If you're writing a **sentence → use inline code**

👉 If you're writing **R commands → use a code chunk**

✨ For your assignment (good practice)

Mix both like this:

1. Clean data (chunk)
2. Analyse (chunk)
3. Explain results (inline + text)

If you want, I can show you a **full sample paragraph (high distinction style)** combining marketing spend + purchase amount + inline code.

I've written this paragraph for my assignment. Could you help me make it clearer and more professional for a data analysis report, but still simple and natural? Please keep it sounding like a normal student, not too formal or AI-like.

Here is my paragraph: Online merchants spend money on tailored marketing to encourage customers to make purchases, but it's not obvious how well this increases revenue. Gaining an understanding of this link facilitates more efficient marketing budget allocation. This study examines data from a recent campaign that involved 150 different clients, each of which had a different marketing budget. The objective is to ascertain whether increased marketing expenditures per customer are linked to increased purchase quantities.

Here's a clearer, more natural version that still sounds like a student but reads more like a proper data analysis report:

Online merchants often spend money on targeted marketin
make purchases, but it is not always clear how effective thi
revenue. Understanding this relationship can help business
budgets more efficiently. This analysis uses data from a rec
customers, where each customer received a different level
is to examine whether higher marketing expenditure per cu
higher purchase amounts.

✨ Why this version works better

- **Simpler wording** (e.g. "not obvious" → "not always cle
- **More natural flow** between sentences
- **Clearer objective** ("examine whether… is associated w
- Uses common data analysis language (like *"associated* too strong/causal)

If you want, I can also help you write the **next paragraph (n** everything flows nicely for your assignment 👍

I've written this paragraph for
like the wording could be clear
revise it so it sounds more pro
for a data analysis report, whil
natural?

Also, please make sure it does
AI-generated text. I want it to
student would realistically writ

Here is my paragraph: The pu
determine whether increased
customer are linked to higher
trends between these two vari
dataset is the main goal of the
to find out whether higher mar
associated with better consum

Here's a smoother and clearer version that keeps a natural student tone:

The purpose of this study is to examine whether higher marketing spend per customer is associated with higher purchase amounts. The analysis focuses on identifying patterns between these two variables across the dataset. In particular, it aims to determine whether increased marketing expenditure is linked to stronger customer spending behaviour.

✨ Why this works better

- Removes repetition (you originally said the same idea 3 times in slightly different ways)
- Keeps wording simple but more precise (*"examine", "identify patterns", "associated with"*)
- Flows logically from general purpose → specific aim

If you want, I can help you make your **intro + objective + method paragraphs flow together like a full report** 👍

I've written this paragraph for my assignment, but I feel like the wording could be clearer. Could you help me revise it so it sounds more professional and appropriate for a data analysis report, while still keeping it simple and natural?

Also, please make sure it doesn't sound too formal or like AI-generated text. I want it to sound like something a student would realistically write.

Here is my paragraph: Customer-level observations from a customised marketing campaign are included in the dataset. Variables like marketing spend, number of visits, purchase amount, and customer segment are included in each row, which represents an individual customer. The information shows the relationship between consumer behaviour and different levels of marketing exposure. Certain variables need to be cleaned up before analysis

since they have missing valu
marketing expenditure and p
investigated using this datas

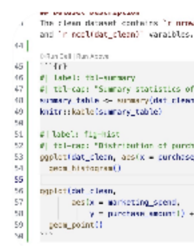
Here's a clearer and more natural revision that still sound

The dataset contains customer-level data from a custom
row represents an individual customer and includes varia
number of visits, purchase amount, and customer segme
explore how customer behaviour may vary with different
Some variables contain missing values and will need to b
This dataset will be used to investigate the relationship b
purchase amount.

✨ Why this works better

- Smoother sentence flow (less repetitive structure)
- Clearer explanation of what each row represents
- Keeps wording simple but appropriate for a report
- Avoids sounding too formal or overly technical

If you want, I can help you connect this with your **previou**
your report reads like one cohesive piece 👍



You're *very close*—just a couple of small issues 👍

❗ Problem 1: Inline code not working

You wrote:

```R
The clean dataset contains `r nrow(dat_clean)` observations and `r ncol(da
```

👉 This is **correct syntax**, BUT it only works if:

- it is **outside a code chunk**
- and you are **knitting the document**

🔧 Fix checklist:

- Make sure this line is **NOT inside** `{r}`
- Click **Knit**, not just "Run"

❗ Problem 2: Typo

```R
varaibles ❌
```

👉 Fix:

```R
variables ✅
```

❗ Problem 3: Your chunk labels are not working

You wrote:

```R
#| label: tbl-summary
#| tbl-cap: "Summary statistics of the cleaned dataset"
```

👉 This syntax is for **Quarto**, not R Markdown.

✅ Fix depending on what you're using

✔ If you are using R Markdown (.Rmd)

Use this instead:

```Markdown
```{r tbl-summary}
summary_table <- summary(dat_clean)
knitr::kable(summary_table, caption = "Summary st
```
```

✔ If you are using Quarto (.qmd)

Then your syntax is correct ✅

BUT make sure your file is actually `.qmd`

❗ Problem 4: Two plots in one chunk (not ideal

Right now you have:

```R
ggplot(...) + geom_histogram()

ggplot(...) + geom_point()
```

👉 Better to separate them:

```Markdown
```{r fig-hist}
ggplot(dat_clean, aes(x = purchase_amount)) +
  geom_histogram()
```

```{r fig-scatter}
ggplot(dat_clean, aes(x = marketing_spend, y = pu
  geom_point()
```
```

✅ Final cleaned version (best practice)

Text (inline code)

Markdown

```
The clean dataset contains `r nrow(dat_clean)` observations and `r ncol(da
```

Summary table

Markdown

```
```{r tbl-summary}
summary_table <- summary(dat_clean)
knitr::kable(summary_table, caption = "Summary statistics of the cleaned d
```
```

Plots

Markdown

```
```{r fig-hist}
ggplot(dat_clean, aes(x = purchase_amount)) +
  geom_histogram()
```

```{r fig-scatter}
ggplot(dat_clean, aes(x = marketing_spend, y = purchase_amount)) +
  geom_point()
```
```

🧠 Key takeaway

- Inline code → **in text, with** `r ...`
- Code chunks → **for running code**
- Don't mix Quarto syntax with R Markdown

If you want, I can help you **write the interpretation for your histogram + scatterplot** (this is where you get easy marks 👀).

Can you explain what i sh
a s ub s e c t i o n that will
You must include the size
such as the number of ob
variable types. You must i
includes i n l i n e R c o d e
variables and observation
Create two other summar
and one as a table. Examp
of certain variables in gro
numeric variables. Make s
summaries are providing a
and are not just arbitrary

This question looks long, but it's actually asking you to
section 👍

🧠 What you need to do (simple versi

Create a **"Data Description" subsection** that includes

✅ 1. Dataset size (with inline code)

You must say:

- number of observations (rows)
- number of variables (columns)
- variable types (e.g. numeric, character)

✅ 2. One table summary

(e.g. summary statistics or grouped counts)

✅ 3. One plot

(e.g. histogram or bar chart)

✍️ How to actually write it (you can follow this exactly)

🔹 Step 1: Write the subsection title

Markdown

```
## Dataset Description
```

🔹 Step 2: Describe dataset size (WITH inline code)

Markdown

```
The clean dataset contains `r nrow(dat_clean)` observations and `r ncol(da
The dataset includes both numeric variables (such as marketing spend, visi
and categorical variables (such as customer segment).
```

👉 ✅ This satisfies:

- inline code requirement
- variable types requirement

🔹 Step 3: Add a table summary

Option A (easiest – recommended)

Markdown

```
```{r}
summary_table <- summary(dat_clean)
knitr::kable(summary_table, caption = "Summary statistics of the cleaned d
```
```

👉 This shows:

- min, max, mean, median
- good for **numeric overview**

Option B (more relevant to your topic 🔥)

Markdown

```
```{r}
library(dplyr)

dat_clean |>
  group_by(segment) |>
  summarise(avg_purchase = mean(purchase_amount
```
```

👉 This is better because:

- links to **customer behaviour**
- not "random" (important for marks)

🔹 Step 4: Add a plot

Option A (you already did this ✅)

Markdown

```
```{r}
ggplot(dat_clean, aes(x = purchase_amount)) +
  geom_histogram()
```
```

Option B (STRONGER for your assignment 🔥)

Markdown

```
```{r}
ggplot(dat_clean, aes(x = marketing_spend, y =
  geom_point()
```
```

👉 This directly shows:
relationship between your main variables

💡 VERY IMPORTANT (for marks)

The question says:

> "not just arbitrary commentaries"

👉 This means:

- Don't just show random plots
- Everything must relate to:
  marketing spend ↔ purchase amount

✅ What a good answer includes

✓ Inline code sentence
✓ Clear variable types
✓ Meaningful table
✓ Meaningful plot

🚀 Pro tip (to get higher marks)

After your table/plot, add **1–2 sentences explanation**, e.g.:

Markdown

```
The summary statistics show that purchase amounts vary widely across custo
The scatter plot suggests a positive relationship between marketing spend
indicating that higher spending may be associated with increased customer
```

🧾 One-line summary

👉 You are basically:

- describing your dataset
- showing 1 table + 1 plot
- and linking everything to your analysis goal

If you want, I can help you **write the exact final version using your actual output** so you can copy-paste into your assignment 👌

I've written this paragraph for my assignment, but I feel like the wording could be clearer. Could you help me revise it so it sounds more professional and appropriate for a data analysis report, while still keeping it simple and natural?

Also, please make sure it d
AI-generated text. I want it
student would realistically v
The dataset was cleaned B
with missing values in the p
guarantees that every data
accurate purchase informa
eliminated from the study s
the accuracy of the finding

Here's a clearer and more natural version that still sound

The dataset was cleaned by removing observations with
amount variable. This ensures that all observations inclu
purchase information. Missing values were excluded bec
accuracy of the results.

✨ Why this works better

- Smoother wording ("eliminating" → "removing")
- Fixes small grammar issue ("By" → "by")
- Less repetitive while keeping the same meaning
- Sounds natural but still appropriate for a report

If you want, I can help you **connect this with your code**
with your reproducible report 👍

I've written this paragraph f
like the wording could be c
revise it so it sounds more
for a data analysis report, v
natural?

Also, please make sure it d
AI-generated text. I want it

student would realistically write. Heres my paragraph: The relationship between marketing spending and purchase amount is shown in @fig-scatter. greater marketing expenditures seem to be positively correlated with greater purchase values. This pattern is supported by the fitted trend line, which implies that higher marketing expenditures are linked to higher consumer spending. There is still significant variance surrounding the trend, though, suggesting that other variables might potentially have an impact on consumer behaviour.

The distribution of purchase amounts among client segments is shown in @fig-box. The Enterprise segment seems to have a higher median purchase amount than the Consumer and SMB segments, despite the fact that all segments show a range of values. This implies that, in addition to marketing expenditure, client segment may also have an impact on purchase behaviour.

Here's a clearer and more natural revision that still sounds like a student:

The relationship between marketing spend and purchase amount is shown in @fig-scatter. There appears to be a positive relationship, where higher marketing spend is generally associated with higher purchase amounts. This is supported by the fitted trend line, which suggests an overall upward pattern. However, there is still a fair amount of variation around the trend, indicating that other factors may also influence customer behaviour.

The distribution of purchase amounts across customer segments is shown in @fig-box. While all segments show a range of values, the Enterprise segment appears to have a higher median purchase amount compared to the Consumer and SMB segments. This suggests that customer segment, in addition to marketing spend, may also play a role in influencing purchase behaviour.

✨ Why this works better

- Fixes grammar (e.g. "greater marketing expenditures..." → smoother phrasing)
- Avoids repeating the same idea too strongly
- Uses cautious wording (*"appears", "suggests"*) → better for data analysis
- Keeps it simple, clear, and realistic for a student

If you want, I can help you add **1–2 stronger analytic**
your research question) to push this into higher mark

this study investigated t
marketing expenditure a
using campaign data. Hi
typically associated with
according to the data, w
correlation between mar
behaviour.

Variation in the data, how
outcomes may also be in
as consumer segment. A
account in conjunction w
generally support the ide
linked to greater purchas
This emphasises how cr
both marketing tactics a
examining consumer beh

Here's a smoother and more natural revision that kee
clear and appropriate for a report:

This study examined the relationship between market
amount using campaign data. The results suggest a p
marketing spend is generally associated with higher c

However, there is noticeable variation in the data, indi
may also be influenced by other factors, such as cust
spend appears to play an important role, it should be
additional factors. Overall, the findings suggest that b
customer characteristics are important when analysin

✨ **Why this works better**

- Smoother flow and less repetition
- More natural phrasing (e.g. "point to a positive correlation" → "suggest a positive relationship")
- Uses cautious, appropriate language for analysis (*"suggest", "appears"*)
- Keeps it simple and realistic—not overly formal

If you want, I can help you **upgrade this into a really strong conclusion (HD level)** by adding a final insight or implication 👀